# StudySpot Final Project Proposal

*Stevens Campus Study Space Finder*

**CS 546 Web Programming I**

**Team Members:** Samantha Cardo, Yifei Du, Alyssa Jones, Lin Qing, and James Schorle

## Project Overview

StudySpot is a Stevens-specific web application that helps students find campus study spaces based on how they plan to work. It combines a directory of campus locations with practical details, community reviews, comments, and time-stamped crowd reports. Students can search for a space, compare study conditions and amenities, save favorites, ask or answer questions, and share recent observations. The application will focus only on Stevens buildings and commonly used campus study areas.

## Need and Value

Students currently rely on memory, word of mouth, or walking between buildings to find an appropriate place to work. A quiet space for individual study may not suit a group meeting, and seating, outlets, accessibility, and noise can vary by location. StudySpot will turn this informal campus knowledge into structured, searchable information so students can make a more informed choice before traveling across campus.

## Data Strategy and Scope

The team will seed MongoDB with approximately 10 to 20 frequently used Stevens study spaces using publicly available campus location information and manual verification. Students will then contribute reviews, comments, new locations, and crowd check-ins. StudySpot will not claim to show live seat availability: every crowd report will display its submission time and will expire after two hours. This approach keeps the initial scope manageable and avoids dependence on an external real-time data source or paid API.

## Core Features

1. **Home Page and Study Space Directory.** The home page will explain StudySpot and highlight recently reviewed or updated spaces. The directory will list available campus locations with a short description, supported study types, key amenities, average rating, and most recent valid crowd status.
2. **Search Filters and Sorting.** Students will be able to search by space or building and filter by individual or group study, noise level, outlet availability, accessibility, and other amenities. Results can be sorted by rating, recent activity, or name.
3. **Study Space Details.** Each location will have a detail page showing its building, room or area, description, study type, amenities, average ratings, recent reviews, and time-stamped crowd reports. The page will also show when its basic information was last updated.

4. **Reviews and Multi-Criteria Ratings.** Registered users will be able to write a review and rate quietness, seating, outlet availability, and overall quality. Users can edit or delete their own reviews, and each space page will display updated averages and the number of submitted ratings.
5. **Submit and Manage Study Spaces.** Users can suggest a Stevens study space that is not already listed by providing its building, room or area, description, study type, and amenities. Before submission, the application will warn the user if another listing already uses the same building and room or area name. Contributors can edit or remove the locations they submit.
6. **Favorites and Saved Spaces.** Students can add or remove study spaces from a personal favorites list and view all saved locations in one place. Their account page will also provide convenient links to the spaces, reviews, and crowd check-ins they have submitted.
7. **Time-Limited Crowd Check-Ins.** Users at a study space can report the current crowd level as Low, Moderate, or Busy. The latest report and submission time will appear on the location page, and reports older than two hours will be marked expired instead of being presented as current information.
8. **Comments and Questions.** Registered users can post comments on study space pages to ask questions, share short observations, and reply to other students. Users can edit or delete their own comments, and each comment or reply will display its author and submission time.
9. **Study Space Comparison.** Users can select two study spaces and compare their study types, amenities, ratings, and most recent valid crowd reports in a side-by-side view.
10. **Crowd Trends.** Each study space can summarize its time-stamped crowd check-ins by day and time to show commonly reported busiest and quietest periods. These trends will be presented as community-reported patterns rather than guaranteed occupancy information.

## Extra Features

1. **Content Reporting and Moderation.** Allow users to report inappropriate, inaccurate, or duplicate content and provide administrators with tools to review and remove flagged submissions.
2. **Temporary Notices.** Allow students to report short-term conditions such as maintenance, closures, or unusual noise.
3. **Interactive Campus Map.** Display study spaces on a Stevens campus map and connect each marker to its detail page.

## GitHub Repository

**Repository URL:** https://github.com/linqing8901-gif/StudySpot